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Total No. of Questions : 13

Total No. of Pages : 02

B.Pharma (Sem–3)

PHYSICAL PHARMACEUTICS-I

Subject Code : BP-302T

M.Code : 93324

Date of Examination : 11-12-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Define briefly :**

   a) Distinguish between solvation and association giving one example each.

   b) What are eutectic mixtures?

   c) What is dielectric constant and its significance?

   d) Define HLB value and its scale.

   e) What are clathrates? Give two examples.

   f) Define 'sodium chloride equivalent'.

   g) What is interfacial tension?

   h) What are polar and semi-polar solvents? Give two examples each.

   i) Define 'freezing point depression'.

   j) Mention the important properties of an amorphous solid.

SECTION-B

2. What is an 'ideal solution'? Differentiate between complete miscibility and partial miscibility phenomena taking suitable examples.

3. Discuss adsorbent - adsorbate interaction at solid-gas interface and explain the adsorption isotherms.

4. Discuss the distribution method for determination of stoichiometric ratio in a complex.

SECTION-C

5. Write a note on influence of buffer capacity and pH on tissues.

6. Summarize the application of diffusion in drug absorption.

7. What are amorphous and crystalline solids? Give examples of polymorphism and highlight the advantages and disadvantages of polymorphic behavior of solids.

8. Write a note on surfactants and their applications in formulation design.

9. How much sodium chloride should be added to 100 ml of a solution containing 1% w/v of apomorphine hydrochloride to be rendered isotonic with serum? (Given : Freezing point depression of 1% w/v solution of apomorphine hydrochloride is 0.08).

10. What are aerosols? Give an account of propellants and their properties desired for use in aerosol dosage forms.

11. Explain hydrophobic interaction and its significance.

12. What is spreading coefficient? Explain the phenomena of spreading at liquid-liquid interface on the basis of work of adhesion and cohesion.

13. Write a note on dissociation constant and its significance.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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